

NETWORKS AND COMPLEXITY

Solution 19-2

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 19.2: Observability and controllability [Lvl. 2]

For the Jacobian from the previous question compute the observability and controllability Gramians

$$\mathbf{P} = - \sum_{m,n} \frac{(\mathbf{w}_m \mathbf{w}_n^\dagger)(\mathbf{v}_m^\dagger \mathbf{v}_n)}{\lambda_n + \lambda_m^\dagger}, \quad \mathbf{Q} = - \sum_{m,n} \frac{(\mathbf{v}_m \mathbf{v}_n^\dagger)(\mathbf{w}_m^\dagger \mathbf{w}_n)}{\lambda_n + \lambda_m^\dagger}$$

and then compute the observability P_{ii} and controllability Q_{ii} of the two nodes.

Solution

From the previous exercise, we know the eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{w}_1^\dagger = (1, 3), \quad (1)$$

$$\mathbf{v}_2 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}, \quad \mathbf{w}_2^\dagger = (0, 1), \quad (2)$$

and the eigenvalues

$$\lambda_1 = -1, \quad \lambda_2 = -2 \quad (3)$$

$$\mathbf{P} = - \sum_{m,n} \frac{(\mathbf{w}_m \mathbf{w}_n^\dagger)(\mathbf{v}_m^\dagger \mathbf{v}_n)}{\lambda_n + \lambda_m^\dagger} \quad (4)$$

$$= \frac{(\mathbf{w}_1 \mathbf{w}_1^\dagger)(\mathbf{v}_1^\dagger \mathbf{v}_1)}{\lambda_1 + \lambda_1^\dagger} + \frac{(\mathbf{w}_1 \mathbf{w}_2^\dagger)(\mathbf{v}_1^\dagger \mathbf{v}_2)}{\lambda_2 + \lambda_1^\dagger} \quad (5)$$

$$+ \frac{(\mathbf{w}_2 \mathbf{w}_1^\dagger)(\mathbf{v}_2^\dagger \mathbf{v}_1)}{\lambda_1 + \lambda_2^\dagger} + \frac{(\mathbf{w}_2 \mathbf{w}_2^\dagger)(\mathbf{v}_2^\dagger \mathbf{v}_2)}{\lambda_2 + \lambda_2^\dagger} \quad (6)$$

Let's consider the numerators of these four terms separately:

$$(w_1 w_1^\dagger)(v_1^\dagger v_1) = \left(\begin{pmatrix} 1 \\ 3 \end{pmatrix} (1, 3) \right) \left((1, 0) \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) \quad (7)$$

$$= \begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix} \cdot 1 \quad (8)$$

$$(w_1 w_2^\dagger)(v_1^\dagger v_2) = \left(\begin{pmatrix} 1 \\ 3 \end{pmatrix} (0, 1) \right) \left((1, 0) \begin{pmatrix} -3 \\ 1 \end{pmatrix} \right) \quad (9)$$

$$= \begin{pmatrix} 0 & 1 \\ 0 & 3 \end{pmatrix} \cdot -3 \quad (10)$$

$$(w_2 w_1^\dagger)(v_2^\dagger v_1) = \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} (1, 3) \right) \left((-3, 1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) \quad (11)$$

$$= \begin{pmatrix} 0 & 0 \\ 1 & 3 \end{pmatrix} \cdot (-3) \quad (12)$$

$$(w_2 w_2^\dagger)(v_2^\dagger v_2) = \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} (0, 1) \right) \left((-3, 1) \begin{pmatrix} -3 \\ 1 \end{pmatrix} \right) \quad (13)$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \cdot (10) \quad (14)$$

Substituting back yields

$$\mathbf{P} = \frac{1}{2} \begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 1 & 3 \end{pmatrix} + \frac{5}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (15)$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \quad (16)$$

So the observability of the first node is 1/2 whereas the observability of the second node is 1. It makes sense that the second node has more impact on the dynamics as it has a strong feedback on the first node.

Now we compute the four numerators that appear in Q

$$(w_1^\dagger w_1)(v_1 v_1^\dagger) = \left((1, 3) \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right) \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} (1, 0) \right) \quad (17)$$

$$= (10) \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (18)$$

$$(w_1^\dagger w_2)(v_1 v_2^\dagger) = \left((1, 3) \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} (-3, 1) \right) \quad (19)$$

$$= (3) \cdot \begin{pmatrix} -3 & 1 \\ 0 & 0 \end{pmatrix} \quad (20)$$

$$(w_2^\dagger w_1)(v_2 v_1^\dagger) = \left((0, 1) \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right) \left(\begin{pmatrix} -3 \\ 1 \end{pmatrix} (1, 0) \right) \quad (21)$$

$$= (3) \cdot \begin{pmatrix} -3 & 0 \\ 1 & 0 \end{pmatrix} \quad (22)$$

$$(w_2^\dagger w_2)(v_2 v_2^\dagger) = \left((0, 1) \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \left(\begin{pmatrix} -3 \\ 1 \end{pmatrix} (-3, 1) \right) \quad (23)$$

$$= (1) \cdot \begin{pmatrix} 9 & -3 \\ -3 & 1 \end{pmatrix} \quad (24)$$

Hence the matrix is

$$\mathbf{Q} = 5 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} -3 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} -3 & 0 \\ 1 & 0 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 9 & -3 \\ -3 & 1 \end{pmatrix} \quad (25)$$

$$= \frac{1}{4} \begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix} \quad (26)$$

This shows that the first node with its controllability of 5/4 is five times more sensitive to perturbations than the second node (controllability 1/4).